

Arithmetic Operation between K-Maps



- *Axioms* or *postulates* of Boolean algebra are a set of logical expressions that we accept without proof and upon which we can build a set of useful theorems.

AND operation

Axiom 1: $0 \cdot 0 = 0$

Axiom 2: $0 \cdot 1 = 0$

Axiom 3: $1 \cdot 0 = 0$

Axiom 4: $1 \cdot 1 = 1$

OR operation

Axiom 5: $0 + 0 = 0$

Axiom 6: $0 + 1 = 1$

Axiom 7: $1 + 0 = 1$

Axiom 8: $1 + 1 = 1$

NOT operation

Axiom 9: $\overline{1} = 0$

Axiom 10: $\overline{0} = 1$

Special Note (Using Boolean Axioms):

• **AND between K-maps** is performed by **multiplying the corresponding cells** of the two maps, following the AND axioms

OR between K-maps is performed by **adding the corresponding cells** of the two maps, following the OR axioms

NOT of a K-map is obtained by **complementing every cell** of the map, following the NOT axioms

All cell-by-cell operations between K-maps are governed strictly by these Boolean axioms.